

UK IT Recruitment Intelligence

How Data Wrangling Transforms Raw Job Listings into
Competitive Advantage for UK Recruitment Agencies

Aisaule Shamil

Who Uses This Analysis?

A UK-based IT Recruitment Agency

IT recruitment agencies operate in a high-velocity market: hundreds of new listings appear daily and disappear within days. Consultants need structured, reliable intelligence to advise candidates on salary expectations, identify high-demand roles, and target the right clients at the right time.

Without clean data, agencies rely on instinct: a costly disadvantage in a data-rich industry.

 Industry

IT Staffing & Recruitment

 Geography

UK (all regions)

 Data Source

Adzuna Job Listings API

 Listings Analysed

2,981 IT job postings

The Challenge Recruitment Agencies Face

Without structured market data, consultants cannot answer three critical questions accurately:

01 Data Overload

Thousands of raw, unstructured job postings across 14 regions: duplicates, missing fields, and inconsistent titles make manual analysis impossible at scale.

02 Salary Uncertainty

56% of listed salaries are Adzuna-predicted, not employer-stated. Consultants quoting uncleaned salary data risk losing both candidates and clients.

03 No Market Map

Which roles are booming? Which regions have the most demand? Without standardized job categories and location data, agencies cannot prioritize consultant focus areas.

From Messy Data to Business-Ready Intelligence

1

Deduplication

Removed 17 duplicate listing IDs: same job counted once. Reliable demand count is the foundation of every insight.

2

Salary Correction

Replaced 20 implausible salary_min values (<£1,000) using regex extraction from description text. Capped outliers via IQR winsorisation.

3

Location Parsing

Parsed location.area from string-formatted lists into separate country and region columns, enabling region-level market analysis.

4

Title Normalisation

Stripped seniority prefixes, location noise, and casing inconsistencies. 2,369 raw titles -> 2,135 normalized -> 9 business categories.

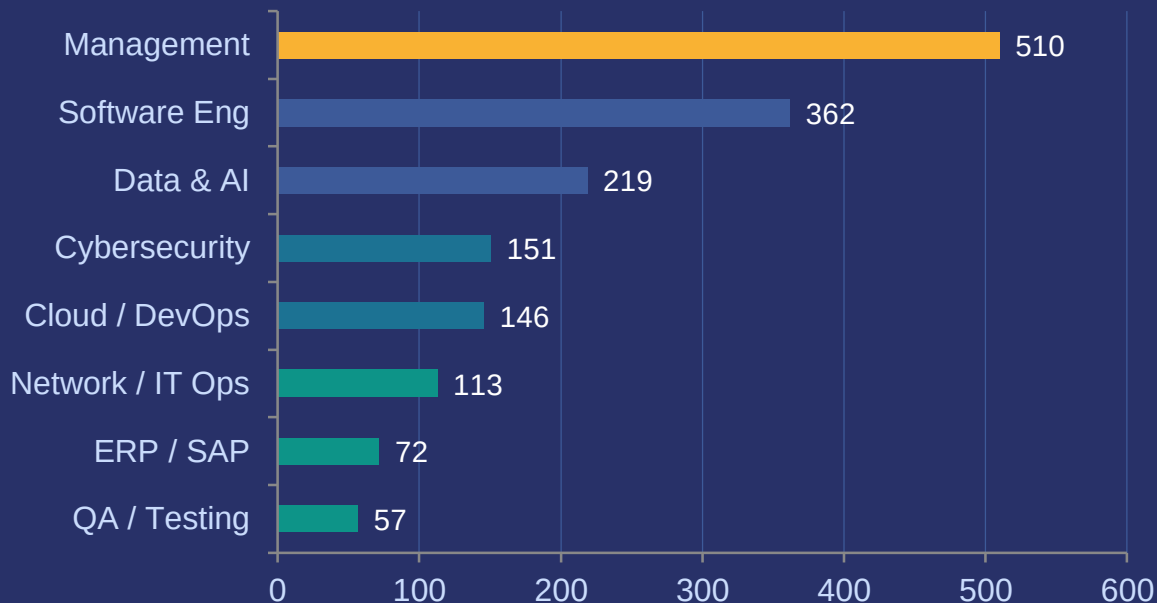
5

API Enrichment

Fetches 500 fresh listings via Adzuna API (26 Apr 2026), added 498 unique new records, keeping the dataset current for live analysis.

Where Is the Demand?

Job listings by category after cleaning and normalising 2,981 UK IT postings



#1 Category

Management

510 active listings

Technical Demand

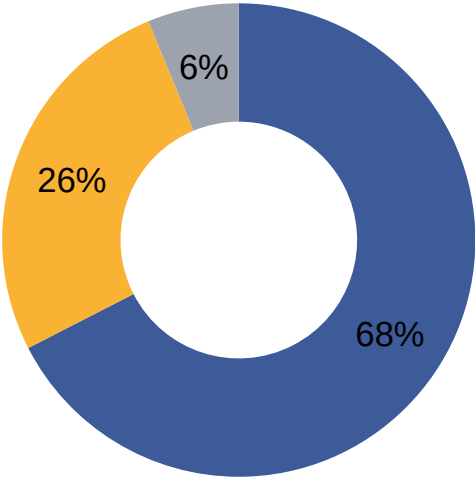
Software Eng leads at 362 listings: highest among pure technical roles, ahead of Data & AI (219) and Cloud/DevOps (146).

The Seniority Gap & Salary Benchmarks

67% of all IT job demand is Mid-Level



The UK IT market is not primarily hiring juniors or seniors, it wants experienced professionals aged 3–7 years in their roles.



■ Mid-Level ■ Senior ■ Junior

Salary Benchmarks by Category (Cleaned Data)

Category	Min (£)	Mid (£)
Management	55,000	82,000
Cloud / DevOps	50,000	75,000
Data & AI	45,000	70,000
Software Eng	42,000	65,000
Cybersecurity	40,000	65,000

What the Recruitment Agency Should Do Next

Specialise Consultants by Top-Demand Category

Software Engineering (362) and Management (510) represent 29% of all cleaned listings. Assign dedicated consultants to these verticals with pre-built candidate pipelines. Reduce time-to-placement by eliminating cross-category context switching.

Lead with Verified Salary Ranges and Not Predicted Ones

56% of Adzuna salary data is algorithmically predicted. Build a client-facing salary guide using only the salary_is_predicted = False subset. This differentiates the agency as a trusted, evidence-based partner in candidate salary negotiations.

Geographic Expansion: Target Mid-England & Scotland

While London dominates, cleaned location data reveals significant volume in Yorkshire, North West England, and Scotland. These regions are under-served by specialist IT recruiters, which is a first-mover opportunity to capture market share outside the capital.

Thank You

Data Wrangling Project Business Application

Aisaule Shamil S1W44H